

Who Gets Left Behind? Digital Exclusion Among Kenyan Women and the Reach of Maternal Health Interventions

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Digital maternal health tools (SMS-based reminders, mHealth applications, and online health portals) are expanding rapidly across sub-Saharan Africa. Yet these tools assume a baseline of digital connectivity that many women do not have. Using data from the Kenya 2022 Demographic and Health Survey ($n = 32,156$ women aged 15–49), this paper examines how wealth, education, geographical residence, and age predict phone ownership and internet use. While 74.7% of Kenyan women own a phone, only 36.9% use the internet, a gap that widens dramatically across socioeconomic strata. Logistic regression reveals that education is the single strongest predictor of internet use: women with no education have odds of internet use roughly 52.6 times lower than women with higher education, even after controlling for wealth, residence, and age. Wealth is the second-largest predictor, with the richest-quintile women nearly 20 times more likely to use the internet than the poorest. Urban residence confers only a modest independent advantage after controlling for socioeconomic composition. Predicted probabilities show that a rural woman with primary education has less than a 5% chance of internet use in the poorest wealth quintile, rising to only 48% in the richest. These findings suggest that digital maternal health interventions systematically exclude the women most at risk and call for a phone-first, education-aware design philosophy.

digital health | maternal health | Kenya | digital divide | mHealth

Maternal mortality remains one of the most persistent global health inequities, with the vast majority of preventable deaths concentrated in sub-Saharan Africa (1, 2). In response, governments, NGOs, and technology companies have invested heavily in digital health tools designed to extend the reach of maternal care beyond traditional clinical settings (3). Kenya has been a particularly active site of mHealth innovation: community health workers use mobile devices to track pregnant patients, SMS systems remind women of antenatal appointments, and mobile-linked platforms deliver nutrition and complication-alert information (3, 4).

Yet a fundamental question underlies these interventions: do the women they are designed to reach actually have access to the digital infrastructure they require? If digital health tools disproportionately reach women who are already wealthier, more educated, and more urban, they risk reinforcing, rather than reducing, maternal health inequities. Understanding the structure of digital exclusion is therefore a precondition for designing interventions that work for the women who need them most.

Significance

Digital maternal health tools are expanding rapidly across sub-Saharan Africa, yet they assume a baseline of connectivity that many women do not have. Using nationally representative data from the Kenya 2022 Demographic and Health Survey ($n = 32,156$), we show that education is the single strongest predictor of internet use, stronger than wealth or geography. Women with no education have odds of internet use roughly 52.6 times lower than women with higher education. A rural woman with primary education has less than a 5% predicted probability of using the internet in the poorest wealth quintile. These findings imply that internet-dependent maternal health interventions, as currently designed, are least reachable by the women who need them most.

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This paper makes two contributions. First, it provides a nationally representative descriptive picture of phone ownership and internet use among Kenyan women aged 15–49 across wealth, education, and residence groups. Second, it uses logistic regression to identify the independent predictors of digital access, disentangling the roles of wealth, education, and geography. Results reveal that education is the dominant gating factor for complex digital tools, that wealth drives an enormous gradient in internet use, and that urban-rural disparities in connectivity are largely explained by socioeconomic composition rather than geography per se.

Background

Digital health interventions in low- and middle-income countries have shown promising results in controlled settings. Mwaisaka et al. (3) found that a combined mHealth and community-based intervention in Kwale County, Kenya improved maternal outcomes for young mothers. Nyatuka et al. (4) argue for a frugal ICT4D framework that prioritises low-cost, locally appropriate digital solutions for maternal care. However, most published mHealth evaluations are conducted in peri-urban settings and disproportionately enrol women with secondary education or above (5–7), precisely the groups least likely to represent Kenya’s highest-risk maternal populations.

Prior literature has documented substantial digital divides in sub-Saharan Africa by wealth, education, and geography, but few studies use nationally representative samples to decompose these predictors simultaneously. The Kenya 2022 DHS provides a rare opportunity to quantify digital access at population scale, linking connectivity variables to the full socioeconomic profile of respondents.

Data and Methods

Data source. This analysis uses the Kenya 2022 Demographic and Health Survey (DHS) Individual Recode (8), a nationally representative cross-sectional household survey of women aged 15–49. The working sample is $n = 32,156$ after listwise deletion of missing values across the six analytic variables.

Variables. Two binary outcomes are examined: *phone ownership* (v169a, owns a mobile telephone, 0/1) and *internet use* (v171a, uses the internet, 0/1). Phone ownership is the hardware prerequisite for mobile-based interventions; internet use is required for app- and web-based tools. Predictors are household *wealth quintile* (v190; 1 = poorest, 5 = richest), *education level* (v106; no education, primary, secondary, higher), *urban/rural residence* (v025), and *age* in years (v012). Table 1 reports the sample distribution across all predictor categories.

Table 1. Sample characteristics. Kenya DHS 2022 women aged 15–49. $N = 32,156$.

Group	<i>n</i>	Phone	Internet
<i>Wealth quintile</i>			
Q1 (Poorest)	7,073	54.3%	5.4%
Q2	5,742	66.8%	17.1%
Q3	6,345	76.4%	31.9%
Q4	7,160	85.4%	53.7%
Q5 (Richest)	5,836	92.5%	79.5%
<i>Education level</i>			
No education	3,836	66.2%	4.9%
Primary	11,807	69.3%	16.8%
Secondary	11,634	73.0%	45.2%
Higher	4,879	98.8%	90.9%
<i>Residence</i>			
Rural	19,770	68.4%	24.3%
Urban	12,386	85.0%	56.9%
<i>Full sample</i>	32,156	74.7%	36.9%

Mean age = 28.6 years (SD not reported; computed from v012).

Analytic strategy. The analysis proceeds in three steps. First, group means of phone ownership and internet use are computed across all predictor categories. Second, two logistic regression models, one per outcome, regress each binary outcome on wealth quintile, education, residence, and age simultaneously, with reference categories of wealth Q1, higher education, and rural residence. Results are reported as odds ratios (ORs) with 95% confidence intervals. Third, predicted probabilities are computed across wealth quintiles holding education fixed at Primary, residence at Rural, and age at the sample mean (28.6 years). All analyses were conducted in Python using statsmodels.

Results

Overall access rates. Across the full sample, 74.7% of women own a mobile phone while only 36.9% use the internet. This 37.8 percentage-point gap indicates that phone ownership substantially overstates effective digital connectivity.

Descriptive patterns. Wealth. Fig. 1 shows that internet use rises steeply from 5.4% in Q1 to 79.5% in Q5, a 74.1 percentage-point gap. Phone ownership also rises with wealth (54.3% to 92.5%), but the gradient is shallower (38.2 pp). In Q1, more than half of women own a phone yet fewer than one in eighteen uses the internet.

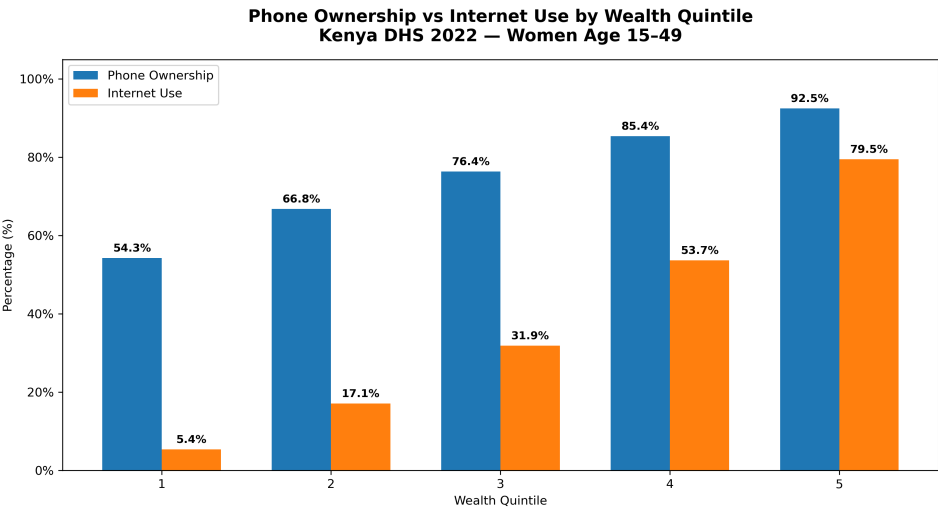


Fig. 1. Phone ownership and internet use by wealth quintile. Q1 = poorest, Q5 = richest. Kenya DHS 2022 (*n* = 32,156).

Education. Fig. 2 shows the steepest gradient of any predictor for internet use: from 4.9% among women with no education to 90.9% among those with higher education, an 86.0 pp gap. For phone ownership the gap is 32.6 pp (66.2% to 98.8%). The jump from secondary to higher education is especially large for internet use (45.2% to 90.9%), suggesting qualitatively different digital engagement at the post-secondary level.

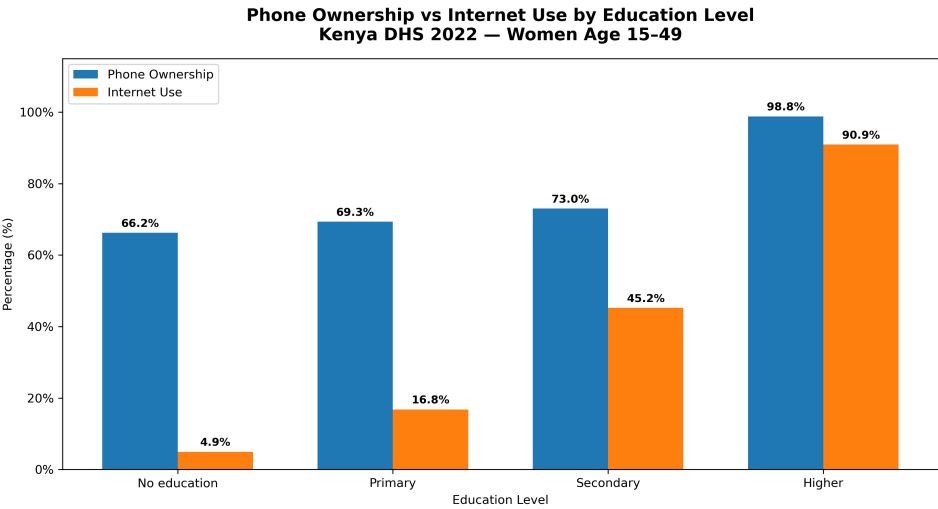


Fig. 2. Phone ownership and internet use by highest education level. Kenya DHS 2022 (*n* = 32,156).

Urban/rural residence. Internet use is 56.9% among urban women versus 24.3% among rural women, a 32.6 pp gap. Phone ownership shows a smaller gap (85.0% vs. 68.4%), and 68.4% of rural women own a phone, indicating a viable foundation for SMS-based outreach.

Compound disadvantage. Fig. 3 shows internet use rates at each wealth-by-education combination. A Q1 woman with no education has a 1.5% internet use rate; a Q5 woman with higher education has 96.8%. Education can partially offset low wealth: a Q1 woman with higher education uses the internet at 54.8%, far above the Q3 average for women with no education (9.8%). Conversely, wealth alone cannot overcome an education deficit: Q5 women with no education use the internet at only 32.4%.

Table 2. Logistic regression odds ratios for phone ownership and internet use. Reference categories: Wealth Q1, Higher education, Rural residence. Kenya DHS 2022 ($n = 32,156$). All ORs significant at $p < 0.05$ except age in the internet use model ($p = 0.293$).

Predictor	Internet Use		Phone Ownership	
	OR	95% CI	OR	95% CI
<i>Wealth (ref: Q1)</i>				
Q2	2.221	(1.947, 2.533)	1.617	(1.480, 1.766)
Q3	4.305	(3.800, 4.878)	2.356	(2.147, 2.585)
Q4	8.673	(7.617, 9.875)	3.619	(3.240, 4.041)
Q5	19.894	(17.176, 23.042)	5.546	(4.792, 6.419)
<i>Education (ref: Higher)</i>				
Secondary	0.132	(0.118, 0.147)	0.078	(0.060, 0.101)
Primary	0.044	(0.039, 0.049)	0.044	(0.033, 0.057)
No education	0.019	(0.016, 0.023)	0.043	(0.032, 0.057)
<i>Residence (ref: Rural)</i>				
Urban	1.103	(1.022, 1.190)	1.161	(1.067, 1.262)
Age (per year)	0.998	(0.995, 1.002)	1.138	(1.133, 1.142)
Pseudo R^2	0.357		0.273	
N	32,156			

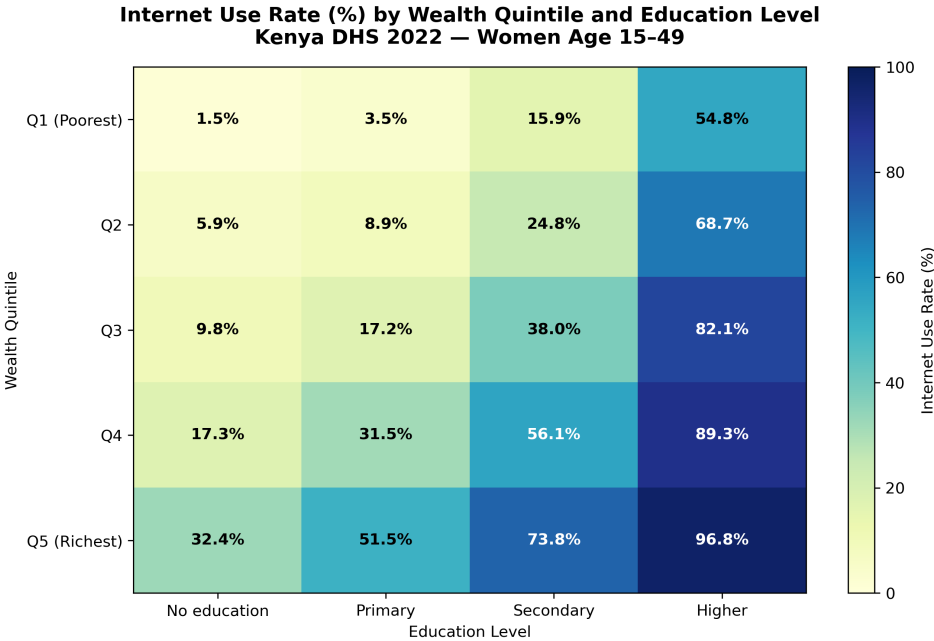


Fig. 3. Internet use rate (%) by wealth quintile (rows) and education level (columns). Kenya DHS 2022 ($n = 32,156$).

Logistic regression. Table 2 reports odds ratios from both models. Fig. 4 visualises these on a log scale.

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Odds Ratios for Digital Access — Kenya DHS 2022
Reference: Wealth Q1, Higher Education, Rural

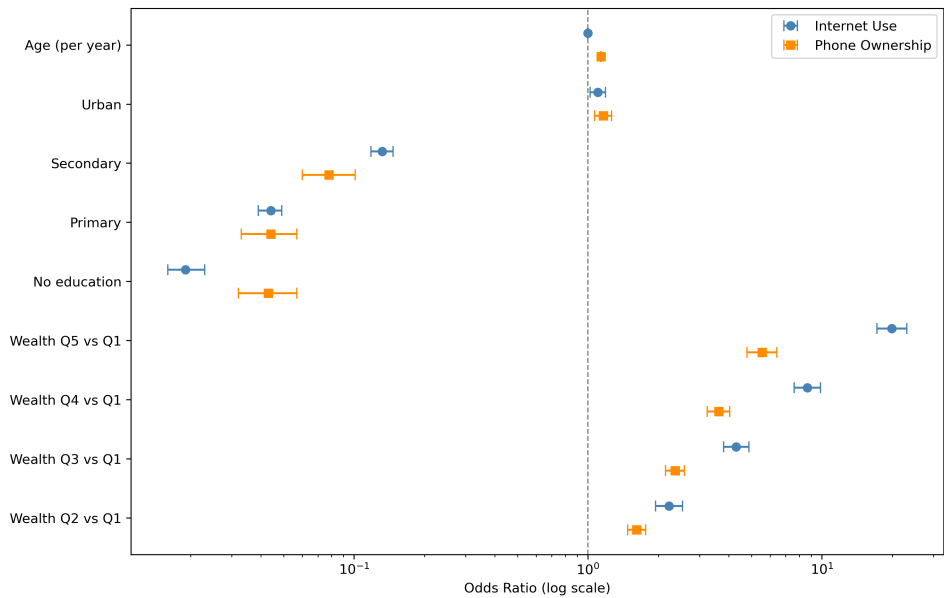


Fig. 4. Forest plot of odds ratios (log scale) for internet use and phone ownership. Dashed line at OR = 1. Reference: Wealth Q1, Higher education, Rural. Kenya DHS 2022 ($n = 32,156$).

Education is the dominant predictor. Women with no education have odds of internet use roughly 52.6 times lower than those with higher education ($OR = 0.019$; 95% CI: 0.016–0.023), exceeding the largest wealth contrast (Q5 vs. Q1: $OR = 19.894$). The education effect is larger for internet use than phone ownership ($OR = 0.043$ for phones), indicating that education shapes the *ability to use* digital services beyond the ability to acquire a device.

Wealth drives a massive gradient. Controlling for education and residence, richest-quintile women have nearly 20 times the odds of internet use versus poorest-quintile women. The gradient is monotonic and accelerating across quintiles (Q2: 2.22; Q3: 4.31; Q4: 8.67; Q5: 19.89). The wealth effect on phone ownership is significant but substantially smaller (Q5 vs. Q1: $OR = 5.55$).

Urban residence confers modest independent advantage. After controlling for wealth and education, urban women have only 10.3% higher odds of internet use than rural women ($OR = 1.103$; $p = 0.012$). This small effect, relative to the large raw urban-rural gap (32.6 pp), implies that most of the observed urban advantage is compositional, driven by urban areas concentrating wealthier, better-educated populations, rather than reflecting infrastructure differences per se.

Age effects diverge across outcomes. Age has a significant positive effect on phone ownership ($OR = 1.138$ per year; $p < 0.001$), consistent with longer lifetime exposure to mobile adoption. By contrast, age has a near-zero and non-significant effect on internet use ($OR = 0.998$; $p = 0.293$), indicating that internet adoption is driven by wealth and education rather than generational patterns.

Predicted probabilities. Table 3 shows predicted probabilities across wealth quintiles for a rural woman with primary education at the sample mean age, a profile representative of a large share of Kenya’s highest-risk maternal population. Even the wealthiest rural woman with only primary education has a predicted internet use probability below 50% (48.2%), while the poorest has just 4.5%. Phone ownership remains high across the distribution (54.8%–87.0%), underscoring the phone-internet gap.

Table 3. Predicted probabilities by wealth quintile. Profile: Rural, Primary education, Mean age (28.6 years). Kenya DHS 2022.

Quintile	Phone	Internet	Gap
Q1 (Poorest)	54.8%	4.5%	50.3 pp
Q2	66.2%	9.4%	56.8 pp
Q3	74.0%	16.8%	57.2 pp
Q4	81.4%	28.9%	52.5 pp
Q5 (Richest)	87.0%	48.2%	38.8 pp

Discussion

Education as the primary bottleneck. The finding that education dominates wealth as an independent predictor of internet use has direct implications for digital health design. Many connectivity interventions assume that reducing

the cost of devices or data will unlock access for underserved women. This analysis suggests that for women with low education, cost is not the binding constraint: digital literacy (the ability to navigate applications and interpret health information online) may be a prerequisite that cost-based interventions alone cannot supply.

Phone ownership as a viable foundation. Despite the connectivity gap documented here, 68.4% of rural women own a phone, and phone ownership is relatively stable across much of the wealth distribution. This points to a viable infrastructure for reaching marginalised women through SMS-based and voice-based interventions that do not require internet access, as demonstrated by Mwaisaka et al. (3). The findings lend quantitative support to a phone-first design philosophy for maternal health outreach.

Compositional vs. infrastructural urban advantage. The finding that the urban-rural gap largely disappears after controlling for wealth and education reframes the policy question. It suggests that equalising rural women's educational attainment and economic resources, rather than simply expanding rural network coverage, may be the more effective lever for closing the connectivity gap.

Limitations. This analysis is cross-sectional and cannot establish causality. Outcome variables are binary self-reports and do not capture frequency or type of digital use. Standard errors do not account for the DHS cluster sampling design and may be slightly underestimated. Subgroup heterogeneity by region or age cohort is not explored.

Conclusion

Digital maternal health tools carry the promise of extending quality care to Kenya's most vulnerable women. This analysis demonstrates that the women who face the highest risks of maternal mortality (those who are poorest, least educated, and most rural) are precisely those with the lowest rates of internet use. Education is the single largest barrier; wealth is the second. Urban-rural gaps are largely explained by socioeconomic composition. For internet-dependent tools to achieve equity, they must either be paired with intensive digital literacy support or radically simplified to reduce the education threshold required. Without deliberate design for digital exclusion, maternal health interventions risk reaching those already advantaged, leaving behind the women who need them most.

Data, Materials, and Software Availability. All analysis code is available at https://github.com/aulanni-kidd/qss20_final_digital_health_access. The underlying data are the Kenya 2022 DHS Individual Recode, available from the DHS Program (<https://dhsprogram.com>) upon registration.

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